Specialized YOLO11 Detection and Selective Prediction for Cafeteria Waste

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Abstract. Waste-sorting assistance requires more than identifying discarded objects: their appropriate disposal streams depend on material composition, contamination, contents, and local collection rules. We examine a prototype comprising four specialized YOLO11 detectors for seven cafeteria-item classes and a rule-based policy that recommends composting, recycling, or manual review. On an historically exposed development cohort of 248 photographs with 924 AI-assisted object annotations, the frozen detectors achieved 87.78% mAP50 and 67.24% mAP50–95. At a uniform confidence threshold of 0.50, micro precision and recall were 93.42% and 78.35%. Class-specific thresholds selected on 186 validation photographs increased precision to 97.03% on validation and 97.22% on the historical cohort, while reducing recall to 71.43% and 68.18%, respectively. Historical-cohort F1 consequently declined from 85.23% to 80.15%. Only six historical-cohort images were captured at the participating school, and a provenance audit identified neither a confirmed independent test cohort nor verified ground-truth disposal-bin labels. The study therefore establishes a reproducible development-set precision–coverage trade-off, not real-world disposal accuracy or reduced waste contamination.

Keywords: cafeteria waste, YOLO11, object detection, selective prediction, waste classification, evaluation provenance.

1 Introduction

At the end of a school lunch period, a tray may contain leftover food, utensils, a drink carton, and several forms of packaging. Recognizing these items is relatively straightforward to formulate as a computer-vision problem; determining where each belongs is not. A carton may be recyclable only after it has been emptied, while packaging that appears compostable may not be accepted by the local collection program. The U.S. Environmental Protection Agency accordingly emphasizes both the need to consult local recycling guidance and the importance of emptying and cleaning containers [1]. Even a correct visual prediction can therefore lead to an inappropriate disposal recommendation.

Object detection provides a natural foundation for an assistance system because it identifies multiple items and their locations within the same photograph. Yet the confidence assigned to a detection expresses the model's assessment of its visual prediction; it does not certify an object's material, condition, or disposal eligibility. A useful system must distinguish recognition from decision-making and preserve a mechanism for withholding recommendations when the available evidence is insufficient. It must also account for objects that are never detected, since excluding uncertain predictions can make reported precision look stronger while leaving more items unaddressed.

This study evaluates a prototype combining four specialized YOLO11 detectors with a deterministic disposal-suggestion policy. Together, the detectors recognize seven item categories; the policy recommends composting or recycling for selected categories and directs others to manual review. We assess the unchanged checkpoints on available development images, investigate the precision–recall consequences of choosing separate confidence thresholds for each class, and audit the provenance of both images and labels. The central question is not whether the system has already solved cafeteria waste sorting, but how reliably its current visual predictions can support a cautious, auditable recommendation process.

1.1 Related Work

Early waste-recognition systems generally treated the task as whole-image classification. Thung and Yang's TrashNet project classified isolated discarded objects into six material-oriented categories [2]. More recently, Single, Iranmanesh, and Raad introduced RealWaste, using photographs collected in a landfill environment to study recognition under less controlled conditions [3]. The contrast matters: a model trained on cleanly presented objects may face a substantially different visual problem when materials are dirty, crumpled, overlapping, or surrounded by unrelated waste.

Other datasets frame waste recognition as instance localization rather than image-level categorization. Proença and Simões introduced TACO, which provides annotations for litter in its surrounding context and supports instance-segmentation research [4]. Bashkirova and colleagues extended this emphasis on difficult scenes with ZeroWaste, a collection of cluttered recycling-conveyor images containing deformable, overlapping objects [5]. Both illustrate why

mixed waste is poorly represented by a single label per photograph. Neither, however, makes local disposal eligibility equivalent to visual class: a detected plastic object can still require information about its material specification, contents, or collection rules before an appropriate bin can be established.

The detection architecture in this study belongs to the YOLO family introduced by Redmon and colleagues [6] and uses the YOLO11 implementations distributed by Ultralytics [8]. We follow the localization-sensitive evaluation conventions associated with Microsoft COCO, including average precision at multiple intersection-over-union thresholds [7]. These metrics summarize recognition across ranked detections, but they do not specify how a particular confidence cutoff behaves in practice. For an assistance system that acts only on selected predictions, fixed-threshold precision and recall provide complementary information.

The decision to withhold uncertain outputs connects this work to selective classification. Geifman and El-Yaniv examined confidence-based rejection [9] and subsequently introduced SelectiveNet, which integrates a rejection option into neural-network training [10]. Our method is deliberately narrower: it changes confidence cutoffs for existing detectors without retraining them or implementing a learned rejector. Higher precision among the retained predictions must therefore be assessed alongside the objects that become undetected or require review.

Finally, the reliability of any claimed improvement depends on evaluation design. WILDS documents the difficulty of transferring learned models across distributions and collection environments [11], while Kapoor and Narayanan describe how leakage can compromise reproducibility in machine-learning research [12]. These concerns are immediate here: images originate from several datasets, parts of the model-selection history remain incomplete, and the nominal historical test cohort was previously consulted during development. We consequently distinguish between observations supported by the available development records and generalization claims that would require a new, sealed evaluation.

2 Data and Annotation Provenance

The maintained dataset contains 2,372 training views, 186 validation photographs, and 248 photographs in the historical test partition. Of the training views, 2,052 are original image entries and 320 are additional augmented views; twenty school photographs account for 340 views when their augmentations are included. Augmentation increases the variety of presentations seen during training without creating independently captured scenes. Because the specialist checkpoints were developed through different historical training and selection procedures, the current manifest should not be interpreted as the exact common training set for all four models.

The images come from public waste datasets and two school submissions. The historical cohort combines 73 school-lunch dataset images, 66 tetra-pak images, 48 trash-detection images, 31 plate images, 20 utensil images, four macworx images, and six photographs collected at the participating school. Thus, only 2.42% of the historical cohort consists of school-origin photographs. The separate dataset named school_lunch is not part of that school collection, and the cohort cannot be treated as a representative field sample from the cafeteria.

We score detections against the frozen, repaired annotation manifests rather than older label files that no longer reflect the accepted evaluation records. The historical cohort contains 924 annotated objects. A previous validation audit documented discrepancies in 109 older on-disk label files, including ten involving bottle annotations. None of these labels was changed for the present editorial revision; the recorded manifests and hashes remain the authoritative evaluation inputs.

Annotation quality also limits the interpretation of the scores. Historical annotations were produced or reviewed with AI assistance, and the available records do not establish comprehensive independent human verification. The school checklist, for example, contains 123 entries with blank human-review fields. Dense food scenes can also make the boundary between adjacent servings ambiguous. This uncertainty is particularly consequential for hinged containers, represented by just two validation annotations and four in the historical cohort.

Table 1. Annotated objects by class and split. Training counts include augmentation views and are not counts of independent photographed objects; validation and historical-test counts use the frozen repaired annotations.

Class	Training-view boxes	Validation boxes	Test boxes
Food	2,501	343	433
Plate	722	110	123
Utensil	457	101	98
Hinged container	185	2	4
Drink carton	972	94	134
Wrapper	316	85	75

Class	Training-view boxes	Validation boxes	Test boxes
Bottle	293	42	57

3 Model Architecture and Inference

Rather than training one model to recognize every cafeteria item, the prototype combines four previously adapted YOLO11 specialists. Each checkpoint is responsible for a defined subset of the seven output categories, as summarized in Table 2. This division reflects the retained development configuration, not a controlled comparison showing that specialization outperforms a unified detector. The checkpoints, class mappings, inference parameters, and disposal rules were recorded in the project's frozen configuration at commit defcc6156a4425daf1ae530757634cc541f3c267. The protected hashes were checked during the research audit so that the reported comparisons refer to unchanged model artifacts.

Table 2. Frozen specialist checkpoints, parameter counts, class ownership, and per-model inference resolution.

Specialist	YOLO11 variant	Parameters	Input size	Owned classes
food-container-carton	yolo11n.yaml	2,590,425	640	food, hinged_container, drink_carton
plate-wrapper	yolo11n.yaml	2,605,126	640	plate, wrapper
utensil	yolo11x.yaml	56,877,241	800	utensil
bottle	yolo11s.yaml	9,428,179	640	bottle

Inference uses Ultralytics 8.4.175, PyTorch 2.14.1, Python 3.12, batch 1, single-pass inference, NMS IoU 0.70, `max_det=300` per specialist, no test-time augmentation and `half=False`. The last argument is deprecated by the installed library but was retained to reproduce the frozen protocol. Box coordinates are normalized XYXY. Native per-specialist NMS is followed by concatenation of the owned classes; no additional cross-specialist deduplication is applied.

The experiments were run in a recorded environment using an Apple M1 MacBook Air with 8 GiB of memory and MPS acceleration. These specifications document the computational setting; inference time, energy use, and throughput were not benchmarked. The four checkpoints also have different training and selection histories, so the system should not be described as the product of one controlled training run.

The prediction receipt saves detections down to confidence 0.001 for AP analysis. Display/sorting emits scores at least 0.25. Scores below 0.50 require review when displayed; higher scores use the unchanged policy:

Recognized class	Suggested decision at confidence >=0.50
Food	Compost
Bottle, drink carton	Recycling, without checking emptiness
Plate, utensil, wrapper, hinged container	Review

The policy is owner-selected, not evidence that filled containers or every detected food/package qualifies for the school's real collection stream. Independent bin labels must follow actual acceptance evidence rather than the detector mapping.

4 Evaluation Methods

For each predicted box and annotated object box, intersection-over-union (IoU) is the area of their overlap divided by the area covered by either box. The primary localization metrics are mAP50, which averages average precision (AP) across the seven classes at IoU 0.50, and mAP50–95, which additionally averages over IoU thresholds from 0.50 to 0.95 in increments of 0.05 [7]. AP is computed over confidence-ranked predictions using the frozen Ultralytics evaluation implementation. It is distinct from precision measured after selecting one confidence threshold.

At a fixed threshold, predicted objects are matched to annotations one-to-one, in descending order of confidence, when their class labels agree and IoU is at least 0.50. An unmatched prediction is a false positive, while an unmatched

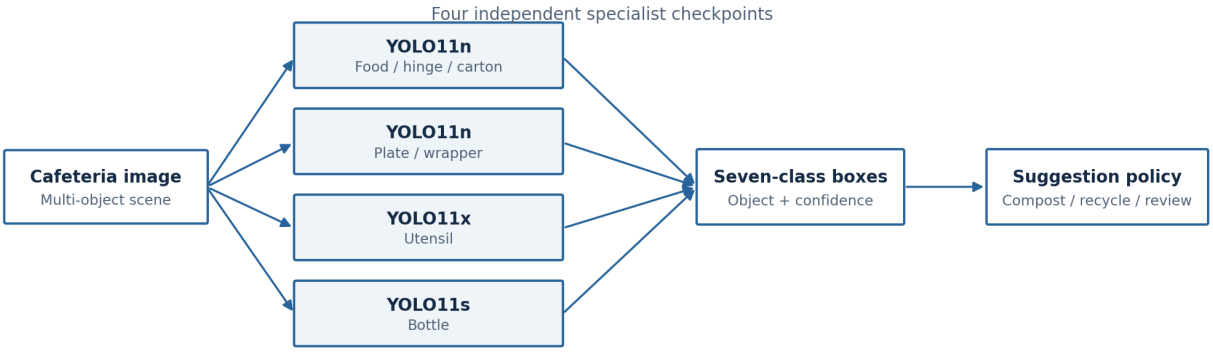


Figure 1: System diagram: four YOLO11 specialists detect seven cafeteria classes before rule-based suggestions or review.

annotation is a false negative. We aggregate true positives (TP), false positives (FP), and false negatives (FN) across all seven classes to report micro-averaged precision, recall, and F1:

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2PR}{P + R}.$$

The suggestion policy is assessed separately because object detection is not itself a disposal-bin decision. First, we sum class-consistent matches for the three classes eligible for an automatic suggestion: food, bottle, and drink carton. We also report a more conservative geometry-first comparison, in which all displayed detections compete for each annotated object regardless of predicted class; the predicted class is checked only after the box is assigned. This reveals cases where overlapping food and packaging predictions would otherwise each appear to be correct. Neither matching scheme establishes the correct disposal stream of an item.

The historical 248-image partition had been examined during earlier development and model-selection decisions. It is therefore described as an exposed development cohort, not a sealed independent test. Thresholds were selected using the 186-image validation cohort and then evaluated unchanged on the exposed historical cohort. In particular, selecting a stricter confidence threshold cannot change the frozen models' mAP, which is computed from the broader ranked detection outputs saved down to confidence 0.001. Test-set reuse and incomplete separation between development and evaluation can lead to optimistic performance estimates [12].

5 Confidence Threshold Selection

We first examined whether a single confidence threshold could reduce false detections across all seven classes. Thresholds from 0.50 to 0.99, in increments of 0.01, and an additional threshold of 0.995 were tested on the 186-image validation set. No global threshold met all three requirements: at least 97% micro precision, at least 60% recall, and at least 100 accepted predictions. The best global threshold satisfying the recall and prediction-count requirements was 0.79, yielding 96.04% precision and 68.73% recall.

We therefore considered a separate threshold for each class. For every class, we enumerated the true and false detections accepted at candidate thresholds on validation. A dynamic program combined these class-level possibilities and selected the policy retaining the most correctly detected objects while meeting the same precision, recall, and prediction-count requirements:

$$\max_{\tau} TP(\tau) \quad \text{subject to} \quad P(\tau) \geq 0.97, \quad R(\tau) \geq 0.60, \quad N_{\text{pred}} \geq 100.$$

Table 3. Confidence thresholds selected exclusively from development validation.

Food	Plate	Utensil	Hinged container	Drink carton	Wrapper	Bottle
0.79	**0.83**	**0.50**	**0.50**	**0.84**	**0.83**	**0.91**

The selected thresholds were applied unchanged to the 248-image historical cohort, without another round of tuning. They were not activated in the default disposal-suggestion pipeline. Since the historical photographs

had already been used during earlier development, this comparison does not provide an independent estimate of performance on new school images.

6 Results

Table 4. Per-class fixed-threshold object-recognition metrics on the exposed 248-photo cohort (confidence 0.50, IoU 0.50); AP uses the full frozen score range. These are descriptive development results, not independent bin classification.

Class	TP / FP / FN	Precision	Recall	F1	AP50	AP50–95
food	327 / 10 / 106	97.03%	75.52%	84.94%	83.91%	55.09%
plate	103 / 15 / 20	87.29%	83.74%	85.48%	87.04%	71.60%
utensil	70 / 7 / 28	90.91%	71.43%	80.00%	86.06%	61.26%
hinged_container	4 / 1 / 0	80.00%	100.00%	88.89%	99.50%	91.17%
drink_carton	129 / 1 / 5	99.23%	96.27%	97.73%	97.55%	74.28%
wrapper	58 / 7 / 17	89.23%	77.33%	82.86%	85.13%	57.64%
bottle	33 / 10 / 24	76.74%	57.89%	66.00%	75.27%	59.62%

The historical development cohort yielded 87.78% mAP50 and 67.24% mAP50–95, compared with 86.36% and 65.25%, respectively, on validation. These are macro-averaged detection scores, so each class contributes to mAP regardless of its sample size. The hinged-container AP50 of 99.50%, for example, reflects only four annotated objects in the historical cohort and is not reliable evidence of generalization for that class. At a fixed 0.50 confidence cutoff, overall historical-cohort micro precision, recall, and F1 were 93.42%, 78.35%, and 85.23%. The underlying per-class validation results appear in `results/sorting-verification.json` and the evaluation report.

Table 5. Automatic-suggestion object proxies, explicitly separate from bin accuracy. Recall uses the 624 food/bottle/carton object labels.

Matching convention	Correct object matches	Precision	Recall	F1
Class-specific	489 of 510 automatic suggestions	95.88%	78.37%	86.24%
Joint geometry-first	483 of 510 automatic suggestions	94.71%	77.40%	85.19%

The six joint-matching conflicts are five food boxes and one bottle box assigned to overlapping wrapper labels. They are object/matching conflicts, not independently proven wrong-bin outcomes. The original 248-image replay was previously verified, and a later fresh inference exactly reproduced its predictions. This finalization reuses those receipts; it does not rerun YOLO or train another candidate.

Table 6. Default sorting-policy outputs at the 0.25 display-confidence floor on the exposed historical development cohort; results are not independently verified bin assignments.

Quantity	Count / denominator	Value
Compost suggestions	337 / 881 emitted detections	38.25%
Recycling suggestions	173 / 881	19.64%
Automatic suggestion rate	510 / 881	57.89%
Manual-review rate among emitted detections	371 / 881	42.11%
Geometry-matched automatic coverage of labeled objects	489 / 924	52.92%
Exact-class correct automatic matches / labeled objects	483 / 924	52.27%

Quantity	Count / denominator	Value
Geometry-matched review / labeled objects	286 / 924	30.95%
Missed objects	149 / 924	16.13%
Unmatched automatic predictions	21 / 510 automatic suggestions	4.12%
Unmatched review predictions	85 / 371 review predictions	22.91%

The default display policy emitted 881 detections at confidence 0.25 or above. Of these, 510 received a compost or recycling suggestion and 371 were marked for review, including predictions below the 0.50 automatic-suggestion cutoff and classes designated as review-only. The 42.11% review fraction applies only to emitted detections; it excludes real objects the detectors missed altogether. Likewise, the 52.27% exact-class automatic-match fraction is a detection proxy, not the proportion of waste placed in the correct bin. Conventional object-independent confidence intervals would be misleading because many objects occur in the same scenes, and the historical images were previously exposed during development.

6.1 Precision–Recall Trade-off After Threshold Selection

Table 7. Overall seven-class micro object recognition at box IoU ≥ 0.50; same source images and labels within each cohort.

Cohort / policy	TP	FP	FN	Precision	Recall	F1
Validation (186), uniform 0.50	638	48	139	**93.00%**	82.11%	87.22%
Validation (186), classwise selected	555	17	222	**97.03%**	71.43%	82.28%
Historical (248), uniform 0.50	724	51	200	**93.42%**	78.35%	85.23%
Historical (248), classwise fixed	630	18	294	**97.22%**	68.18%	80.15%

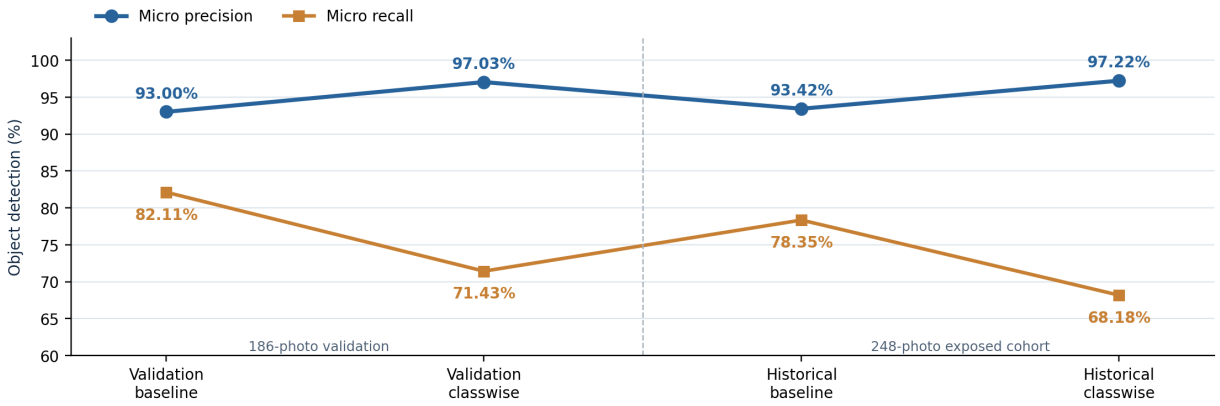


Figure 2: Precision–recall trade-off for default and validation-selected classwise confidence thresholds, measured on exposed development cohorts.

On the historical cohort, the class-specific policy reduced false detections from 51 to 18 but also rejected 94 previously correct detections, decreasing true positives from 724 to 630. Accepted detections fell from 775 to 648. Precision increased by 3.80 percentage points, but recall fell by 10.17 points and F1 decreased from 85.23% to 80.15%. A similar F1 reduction occurred on validation (87.22% to 82.28%). Thus, the policy achieved its narrower objective of reducing false-positive detections; it did not improve the detector's overall F1 or mAP. The historical cohort was not used for a second round of threshold selection.

7 Independence and Annotation Audit

A reliable evaluation requires photographs that have not influenced training, threshold selection, or earlier model decisions. The project's local inventory was therefore screened for duplicate files, overlapping capture groups, and recorded prior inference. The audit covered 102,806 readable image paths, corresponding to 67,422 distinct filesystem inodes and 14,379 byte-distinct images. Within the 9,468 candidate image identities, 3,842 were flagged for possible exposure, group overlap, or duplication; 5,482 had unresolved historical provenance; and 144 original HEIC files could not be visually decoded in the audit environment. No image was *confirmed* independent under the available evidence (Table 8). This does not prove that every candidate was used for model fitting.

The audit compared SHA-256 file identities and decoded-pixel hashes and screened potential near-duplicates using rotation/reflection-invariant comparisons of 64-bit difference hashes against 4,751 images from recorded training directories or prepared manifests. A Hamming-distance threshold of five was used for screening. These tests are deliberately conservative: membership in a training directory is not proof of optimizer exposure, while the absence of a perceptual match cannot rule out related crops, viewpoints, or capture sessions.

Table 8. Provenance status of the 9,468 candidate image identities. The categories reflect what could be established from available records, not a definitive partition of images actually used for training.

Candidate status	Image identities
Exposure, group-overlap, or duplicate flag	3,842
Unresolved historical provenance	5,482
Original HEIC visual decoding unavailable	144
Confirmed independent	0

The school-specific history also prevents a prospective test claim. All 123 photographs in the earlier school submission and all 21 in the later submission shared recorded training or development capture groups. Four images from the October 29 submission entered the training collection, so that capture day was excluded from consideration as an independent test source. Earlier labels such as "sealed test" are superseded by subsequent prediction receipts and model-use records. Several audit leads overlap and must not be added together as unique training exposures.

Annotation provenance is a separate limitation. The metadata audit searched 2,398 JSON files and additional CSV, XML, and YAML files but found no confirmed set of independently human-adjudicated, per-object disposal-bin labels. A historical object-review flag is not a substitute for an actual material and disposal-eligibility annotation. The audit also noted 42 unreadable image aliases and five malformed or inaccessible JSON records, which were retained rather than silently repaired. Full provenance diagnostics and hashes are preserved in `results/research-readiness.json` and the external audit artifacts.

Table 9. Real-world sorting metrics that cannot be computed from the current development annotations.

Metric	Status	Evidence required
Precision, recall, and F1 for actual disposal bins	Not measured	Independent images and adjudicated bin labels
Correct-bin accuracy	Not measured	Verified local disposal eligibility
Automatic coverage of independently labeled objects	Not measured	Complete, independent object annotations
Manual-review burden on new cafeteria photographs	Not measured	A sealed school cohort

The lack of independently verified bin outcomes is not equivalent to a zero score. It means that the necessary ground truth does not yet exist in the audited records.

8 Error Analysis and Exploratory Stress Test

The diagnostic decomposition uses frozen object labels and predictions, without editing annotations or changing thresholds. A false-negative label is classified as below threshold if a same-class box has IoU>=0.50 but score<0.50; a localization candidate has same-class IoU>=0.10 but <0.50; otherwise it has no same-class overlap at that cutoff and saved confidence floor. These are competing hypotheses, not human-confirmed causes. A box can overlap other objects or depend on uncertain serving definitions. Background-or-missing-annotation likewise does not distinguish model hallucination from an incomplete label set.

Table 10. Frozen historical-cohort object-detection error diagnostics for food and bottle at the default 0.50 confidence cutoff. Error types are algorithmic hypotheses, not human-confirmed causes.

Diagnostic category	Food	Bottle
Correct class matches at automatic threshold	327	33
FN: below automatic threshold	53	19
FN: localization candidate	48	1
FN: no same-class overlap	5	4
FP: localization candidate	2	0
FP: duplicate/competing detection	1	0
FP: background or missing annotation	7	9
FP: other-class overlap/annotation ambiguity	0	1

Errors were concentrated in a small number of difficult scenes. One buffet photograph accounted for 52 of the 106 missed food objects (49.06%), while a single outdoor-trash photograph accounted for 10 of the 24 missed bottles (41.67%). Multiple errors in one scene should not be interpreted as independent failures across many photographs. Both scenes remained in the reported results.

AI visual inspection covered 12 deterministically selected error cases from 10 unique photographs; it is not a random audit or a human-label correction. Examples suggest a reusable-looking sports bottle outside the disposable-bottle scope, a possible missing bottle annotation, small cylindrical drink-package class ambiguity, dense buffet scale/serving-granularity problems, and sealed food packaging detected as food. Actual material, contents and bin acceptance cannot be settled from these previews. No labels or sorting rules were changed.

A separate, previously completed bottle-specialist experiment tested one additional epoch of classification-head training. Validation AP50 declined from 82.47% to 77.95%, and AP50–95 fell from 62.00% to 59.18%. Because the candidate failed the predefined validation criterion, it was rejected without being activated or evaluated on the historical test cohort. The frozen system reported here was not retrained during manuscript finalization.

8.1 Exploratory Public-Image Stress Test

After the research freeze, the unchanged four detectors were run on 44 source-screened public photographs (nine bottle-source, six carton-source, 15 utensil-source, 12 hinged-container-source and two wrapper-source images). There were 149 displayed predictions at the 0.25 confidence floor: 12 compost suggestions, 49 recycling suggestions and 88 manual-review outcomes. Three photos had no displayed detections. These counts describe emitted predictions, not verified outcomes: there are no human-verified object boxes, documented capture-day independence, or school bin-eligibility labels. Accordingly, no **human-verified detection mAP, correct-bin accuracy, or sorting precision** is available for the complete 44-image cohort. The separate 41-image approximate-label diagnostic below is not a substitute for that missing evaluation.

A preliminary image-level *source-hint* cross-check found at least one matching model class on 9/9 bottle-source, 5/6 carton-source, 12/15 utensil-source, 4/12 hinged-container-source and 0/2 wrapper-source photos. These fractions are **not recall**: source names are not exhaustive human object labels, and some images contain multiple target items. Visual inspection suggests repeated plate/hinged-container confusion on open foam food boxes and missing wrapper detections. A foil meal tray was detected as `drink_carton` at confidence 0.617, triggering an automatic recycling suggestion under the frozen user-selected policy. This is a plausible wrong-object automatic action; actual local bin eligibility cannot be concluded from the photo alone. A carton photo with three cartons yielded four carton boxes and another carton image yielded none. Crowded bottle images also exhibited overlapping boxes and cross-specialist labels.

A separate provisional comparison scored 41 of these 44 public-source photographs using approximate AI-generated object labels. The unchanged V1 detectors obtained 46.25% mAP50 and 21.16% mAP50–95; a developmental hinged-

container V3 candidate obtained 44.78% mAP50 and 21.47% mAP50–95. These numbers are not comparable to a sealed test benchmark: the annotations were not independently human-verified, three crowded bottle photographs were excluded from the scoring subset, and the entire source set had already been inspected during development. The V3 candidate was not adopted as the frozen system. This comparison is included only to document an exploratory stress test, not to claim a measured out-of-distribution performance change.

These observations describe possible failure modes, not independently measured detection or disposal accuracy, and were not used to authorize a new training run. Individual photo references and limitations are documented in the visual audit. These public images may be out-of-distribution relative to actual cafeteria disposal scenes, and original source rights require review before reproducing them in a publication.

9 Limitations and Future Evaluation

The main limitation of this study is that the reported benchmark images are development data, not a sealed holdout. As demonstrated in research on distribution shift and leakage, performance on reused or mismatched source data may not transfer to the intended deployment setting [11, 12]. Earlier inference, model selection, and related capture groups prevent an independent generalization claim. The object annotations were also AI-assisted, with limited evidence of human verification; in particular, the hinged-container class has only two validation objects and four historical-test objects. Food-serving boundaries and visually similar packaging introduce further annotation uncertainty. The existing audit reduces uncertainty about what data and model versions were used, but incomplete historical records mean that the absence of a detected duplicate cannot establish independence.

A second limitation concerns the distinction between an object class and a disposal decision. The current rule layer does not inspect whether a bottle or carton is empty, whether food is contaminated by packaging, or whether a specific material is accepted by the school's collection program. It cannot automatically assign landfill disposal, and requesting manual review is an abstention rather than a correct sorting outcome. The results therefore cannot quantify correct-bin accuracy or reductions in waste contamination. Latency, throughput, energy use, and user interaction during a real cafeteria deployment have also not been measured. Source-license and attribution records must be checked before any external image redistribution.

The next evaluation should use newly collected cafeteria photographs from capture sessions excluded from all training, threshold selection, and prior inspection. Inclusion criteria and capture-group membership should be fixed before model inference. Annotators who have not seen the model's predictions should mark each relevant object's bounding box and actual disposal eligibility using documented local collection rules and, when necessary, material and contents information unavailable from the photograph. Uncertain cases should be adjudicated or recorded as unscorable under a predefined protocol. On that sealed cohort, the unchanged system can then be evaluated for object detection, correct-bin precision and recall, prediction coverage, and manual-review burden, with uncertainty estimated across independent capture groups rather than treating multiple objects in one photograph as unrelated observations. For each disposal stream, an automatic wrong-bin recommendation should count against the predicted stream; a reviewed or missed item should not be credited as a successful automatic disposal decision. Reporting only accuracy among accepted decisions would conceal the effect of abstention, so the study should also report the fraction of *all* eligible labeled items assigned correctly without review.

10 Conclusion

The four-specialist YOLO11 system achieved 87.78% mAP50 and 67.24% mAP50–95 on 248 historically exposed development photographs. Selecting class-specific thresholds on validation increased micro object-detection precision to 97.03% on that cohort and 97.22% on the exposed historical cohort, but reduced recall and F1. This result demonstrates a controllable precision–coverage trade-off, not a general improvement in recognition or a verified ability to direct cafeteria waste to the correct disposal stream. The next decisive experiment is evaluation of the frozen detector and disposal policy on new, independently annotated school photographs with ground-truth bin eligibility.

11 Reproducibility and Data Availability

The retained implementation, experiment configuration, and summary results are maintained in the GitHub repository. The paper's frozen configuration, `configs/research-freeze.json`, records the four checkpoint hashes, class mappings, preprocessing settings, and active disposal rules. Existing benchmark outputs are in

`results/sorting-verification.json`, with provenance and annotation limitations in `results/research-readiness.json`. The corresponding procedures are documented in `docs/sorting-evaluation.md`.

The underlying images, checkpoint binaries, and detailed local audit artifacts are stored separately from Git in `~/Desktop/yolo-cafeteria-research-assets/`. In particular, the combined dataset manifest and the repaired validation and historical-test annotations are external assets. Source URLs and historical licensing declarations are recorded in `data/datasets.csv` and `data/manifests/` within that asset collection, but licenses and redistribution rights have not been independently verified. The GitHub source alone is therefore insufficient to reproduce native inference, and no public release of the collected school photographs is implied.

The original audit implementation is retained in `scripts/research_audit.py`. With the external asset environment present, its file inventory can be reproduced by executing:

```
../yolo-cafeteria-research-assets/venv/bin/python -B scripts/research_audit.py --output /tmp/cafeteria-asse
../yolo-cafeteria-research-assets/venv/bin/python -B -m unittest discover -s tests -v
```

These commands audit retained files and run software tests; they do not retrain YOLO or establish independent sorting accuracy. A future bin-label evaluation should use `scripts/evaluate_sorting.py` with a new, independently verified disposal-eligibility annotation file and the frozen prediction policy.

12 References

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